

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 4th Year 1st Term Examination-2020
Leather Products Engineering III
LE 4109

Time: 1.5 Hours

Full Marks: 120

- N.B. i) Answer any **TWO** questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

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|------|--|----|
| 1(a) | Mention the acceptance criteria for garments leather. Calculate the consumption of leather for medium size blouson jacket. | 12 |
| 1(b) | Mention the significance of "Grading" and "Enlargement". | 06 |
| 1(c) | Why front neck curve more concave than that of back neck curve? | 06 |
| 1(d) | Compare between kimono and reglan sleeve. | 06 |
| 2(a) | Cite the functions of pleat and compare among knife pleat, inverted pleat and box pleat. | 12 |
| 2(b) | How can you manipulate dart in skirt manufacturing? Explain. | 10 |
| 2(c) | Determine waist readii if the given skirt waist girth 63cm and length 52.4cm. | 08 |
| 3(a) | Distinguish between piping pocket and welt pocket. | 05 |
| 3(b) | How clothes communicate our intangible language from culture to culture and what clothes are playing role for that? | 10 |
| 3(c) | Mention popular clothing for men's and women's in two distinguish ancient era like Rome and Renaissance. | 08 |
| 3(d) | Construct a classic four panel circular skirt. | 07 |

SECTION-B

- | | | |
|------|--|----|
| 4(a) | Mention the design parameters of a waistcoat. Show the complete set of patterns of your designed waistcoat. | 10 |
| 4(b) | Categorize waistcoat. Distinguish among waistcoat, vest and the blazer. | 10 |
| 4(c) | State the speciality of waistcoat. What type of leather you will select for fashionable waistcoat fabrication? And why? | 10 |
| 5(a) | Cite the style lines to construct the framework of a trouser and give a brief description of the individual lines. | 09 |
| 5(b) | State the back preparation of a trouser with a neat sketch. Analyze the fabrication technique of front and back assembling of a trouser. | 16 |
| 5(c) | If a gentleman has a waist of 36 inches, calculate his approximate chest. | 05 |
| 6(a) | Why and how do you develop an inimitable & memorable brand identity for your own leather products industry? | 10 |

- 6(b) Why wrap over is the first choice as a part of ladies' costumes in the travelling? What are the most commonly used fasteners in leather goods manufacturing? State in brief. 10
- 6(c) If a Man's dummy waist is 35". Calculate his chest and half across back. 05
- 6(d) How do accessories differ from fashion accessories and trimmings? 05

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 4th Year 1st Examination-2020
Environmental Science & Engineering
LE 4121

Time: 1.5 Hours

Full Marks: 120

- N.B. i) Answer any two questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

- 1(a) What are the different types of ecosystems? Explain their function and structure. 12
 1(b) Distinguish between food chain and food web. 08
 1(c) A liming water sample has total alkalinity of 574 mg /L as CaCO_3 and the pH of the sample is 11.2. Now calculate the hydroxide, bicarbonate and carbonate alkalinity of the sample. 10
- 2(a) Write briefly how industrial activities can introduce different pollutants into the water body? 10
 2(b) Differentiate between point and nonpoint sources of water pollution. 09
 2(c) A water has the following analysis: CO_3^{2-} : 7 mg/L, Fe^{2+} : 9 mg/L, K^+ : 18 mg/L Cl^- : 30 mg/L, NO_3^- : 6 mg/L, Sr^{2+} : 12 mg/L, Ca^{2+} : 4 mg/L, SO_4^{2-} : 3 mg/L. Now calculate the total hardness, carbonate hardness and non carbonate hardness. 11
- 3(a) Write the commonly used laboratory method to measure organic matter in wastewater. 07
 3(b) In a BOD determination 6 mL of wastewater are mixed with 294 mL of diluting water containing 8.6 mg/L of dissolved oxygen. After 5 days incubation at 20°C the dissolved oxygen content of the mixture is 5.4 mg/L. Calculate the BOD of the wastewater. Assume that the initial dissolved oxygen of wastewater is zero. 13
 3(c) Which form of arsenic is most toxic? Sketch a process flow diagram representing arsenic treatment of groundwater. 10

SECTION-B

- 4(a) How are ecotoxicology, ecology, toxicology and toxic substances interlinked with each other? 08
 4(b) Mention five toxic chemical agents used in the leather manufacturing. Including their uses as well as threats to human organs. 10
 4(c) Why are tannery industries regarded as hazardous waste generators? Explain with example. 12
- 5(a) Mr. Bashar is an experienced person in the footwear sector and is willing to establish a BD shoe factory near the Rupsha river in Khulna with an initial investment of Tk 15 lakh capital. He needs an ECC from the department of environment. Make an screening and predict what types of information and analysis will be required for BD shoe Ltd. To get an ECC. 12
 5(b) Define EIA. Briefly discuss the EIA process with flow diagram. 08
 5(c) Draw a tentative value function curve for employment opportunities and find the environmental quality value for 150 jobs and 850 jobs. 10

- 6(a) What do you mean by impact prediction? How are impact prediction, baseline study, and scoping strongly connected with each other? Explain. 13
- 6(b) Following table illustrates the comparative results of two hypothetical alternative tannery arrangements. Find which site is preferable according to scaling and wighting checklist method. Here, environmental factors to be indexed and weighted from 1 to 10. 17

Table

Envirenmental factors	Units	Alternative one	Alternative two
New extended area	Sq. ft	1200	1500
Enhanced production	Sq.ft/ month	10,000	13,000
New job opportunities	Numbers	50	60
Water quality degradation	Relative Magnitude	1X	3X
Socio economic condition	Relative Magnitude	1X	4X
Health and safety degradation	Relative Magnitude	1X	2X